



Fig. 1: Quantum threat timeline



Fig. 2: Quantum investments

2. Make sure the product of p and q (let's call it n) is of desired bit length, say 1024.
3. Let's compute the value of n and $\phi = (p-1)(q-1)$
4. Let's select a proper integer e between 1 and ϕ
5. Secret component derivation: d , again between 1 and ϕ , $\rightarrow ed = 1 \mod \phi$
6. The public key part is (n, e) and private part is (d, p, q) .
7. d, p, q and ϕ are secrets.

Quantum cybersecurity perspective

The prime factorisation we discussed above, when tried out in classical computer, takes a very long time to be computed. This makes it practically invincible, so we feel secure that our classical security model is intact and in place. However, that is not the case for quantum computing paradigm. Fig. 3 describes a mind-blowing comparison of prime factorisation across classical

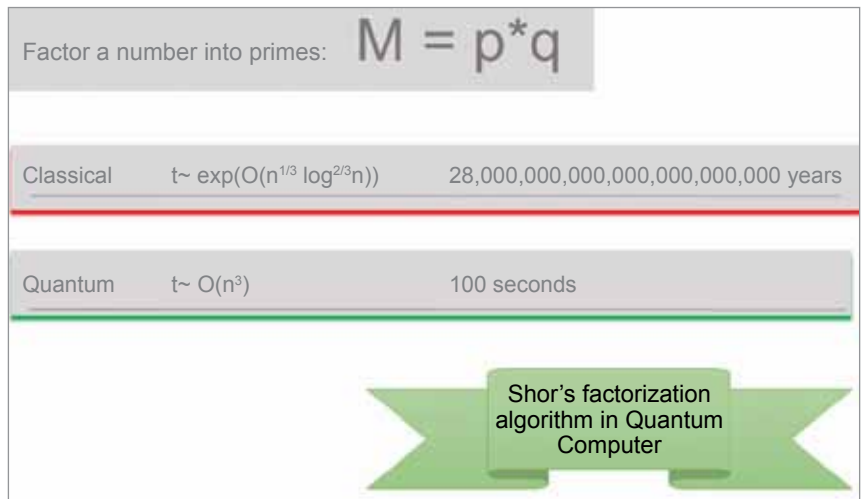


Fig. 3: Computational time comparison between classical and quantum computing for prime factorisation (Shor's algorithm)

and quantum computing paradigms. The corresponding quantum computing prime factorisation algorithm is known as Shor's Algorithm.

The above figure vividly depicts the potential risks of proliferation of quantum computing in the cybersecurity paradigm. To understand the contour of cybersecurity in the quantum age we need to first define what is quantum safe security? Quantum safe security is often called post quantum security as well. Like any modelling of a security problem, let us start with the worst.

Let us say the honest party has only the classical computer and the adversary is quantum powered. Let us assume the adversary's quantum power does not necessarily exist today, it can be futuristic also—to make the problem statement interesting. In the classical world we would like to build a security mechanism that cannot be compromised, even by the quantum powered adversary. Winning in such unfavorable condition is called post quantum security. This is one area attracting intense research interest today.

Let us look at the remediation—how to achieve such a post quantum security paradigm.

To understand this seemingly unbelievable world of post quantum security we need to understand what sort of problems a quantum computer can potentially solve. Can it solve any NP complete (hardest) problem? No, it cannot. Fig. 4 depicts the Bounder Error Polynomial (BQP) time, which is

the problem zone quantum computers can solve in polynomial time.

As we can see in Fig. 4, there still exists an unsolvable zone which is quantum-hard also. It means this zone is out of reach for classical as well as quantum computers. Now, intuitively, if we can devise our security primitives and constructs in that zone (the green colour zone in Fig. 4), then we are safe. Well, we are right, but partially. Having security primitives in the green zone in Fig. 4 is necessary, but not sufficient to prove the quantum safety of the solution.

There are believed to be some security techniques available today which are promising in post quantum order, such as secret-key cryptography. However, the quantum property of the quantum computer, on the other hand, raises additional security attack surface as well. One such example is Superposition Attack. Stating in simple terms, superposition is a quantum property where a qubit (a bit in quantum computer is called qubit) can possess 0, 1, or a mix of 0 and 1 values. Once you measure the qubit, it collapses to either classical 0 or classical 1. Now an adversary, powered by the quantum computer oracle, can potentially learn the superposition ciphertexts of some plain text and decipher the superposition using another algorithm (without directly measuring the superposition ciphertext) to acquire knowledge about the cryptosystem. That's it.

One can, of course, argue that to pose such a threat to our contempo-